

Campus parking and exit carpool coordination

WORKSHOP 3: Robust System Design and Project Management

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Course: Systems Analysis & Design

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1. Executive Summary

This document presents the enhanced conceptual design and project planning framework for the Campus Parking and Exit Carpool Coordination, an information system to manage parking capacity at the engineering campus of Universidad Distrital and coordinate shared trips from the campus to key city zones during peak departure windows. The project showered on Workshop 1 and 2 which showed severe parking saturation, daily overflow of around 90 motorcycles to surrounding streets, access bottlenecks at the Calle 40 basement, and a strong dependence on informal coordination through messaging apps.

The initial was originally framed as Sustainable Transportation Coordination, combining inter-campus carpooling and bike-sharing, but feasibility analysis revealed that an institutional bike-sharing is not realistic in the short term given infrastructure, safety, and operational constraints. The scope was therefore refined and the system renamed Campus Parking and Exit Carpool Coordinator, focusing on conceptual design for parking inventory and reservation management, policy-based prioritization of shared vehicles, and exit-time carpool matching towards city zones

Workshop 3 does not involve software implementation; instead, it delivers a robust conceptual architecture, a risk and quality framework aligned with ISO 31000 and ISO/IEC 25010, and a project management plan that can guide a future development effort.

2. System Evolution

Workshop 1 – System Analysis.

The team analysed transport and parking conditions at the engineering campuses using field observations, parking measurements, process documentation and a survey of 51 users. Results showed 55% of respondents arriving before 8:00 a.m., recurrent saturation of the Calle 40 basement, overflow of about 90 motorcycles onto public streets, and access control based on a mix of digital registration and manual security decisions. Users relied heavily on public transport and informal car-sharing coordinated via messaging apps, motivating the need for institutional coordination mechanisms.

Workshop 2 – Initial Architecture.

Based on these findings, Workshop 2 defined a layered architecture with a student app and security dashboard, core services for ride matching and status updates, an integration layer with vehicle registration and user authentication, and a data and monitoring layer for trips and parking records. This blueprint still reflected the original Sustainable Transportation Coordinator concept (inter-campus trips and potential bike-sharing support), but already incorporated scalability, reliability and data privacy requirements for future operation.

Scope Refinement and Workshop 3.

Between Workshops 2 and 3, the team concluded that short-term deployment of bike-sharing is not feasible, while parking saturation and unmanaged exit-time flows are urgent and implementable priorities. The project was therefore renamed Campus Parking and Exit Carpool Coordinator and re-scoped to focus on conceptual design for parking

management and exit-time carpooling, plus risk, quality and project management structures, without developing code in this phase.

3. Architecture Refinement

3.1 Conceptual Architecture Overview

The Campus Parking and Exit Carpool Coordinator is organised as a layered conceptual architecture that separates user interfaces, application services, institutional integrations and data management to support robustness, modularity and scalability. The goal of this Workshop 3 design is to define clear responsibilities and interfaces between components, without committing to a specific technology stack or implementing code.

At a high level, the architecture includes:

- Client Layer – web interfaces for students and security staff.
- Application Layer – conceptual services for authentication, parking management, exit-time carpool coordination and notifications.
- Integration Layer – logical adapters to institutional user and vehicle registries.
- Data and Monitoring Layer – conceptual repositories for users, vehicles, parking states, carpool offers and audit logs, plus monitoring indicators.

This structure refines the layered blueprint introduced in Workshop 2, aligning it with the updated scope focused on parking management and exit-time carpooling.

3.2 Conceptual Web Interfaces

The Client Layer defines two main web interfaces at a conceptual level:

- Student Web/Mobile Interface.

The student interface groups three core views:

1. a dashboard showing current parking status, active reservations and upcoming exit-time carpool matches,
2. a parking module to register vehicles, request and cancel parking slots, and view assigned spaces, and
3. an exit-time carpool module where drivers publish offers and passengers join shared trips from the campus to predefined city zones within selected departure windows. Interaction flows are kept simple: authenticate with institutional credentials, manage parking and carpool actions from the same navigation bar, and receive confirmation and update messages for each operation.

- Security and Operations Dashboard.

A separate web dashboard is defined for security staff and campus operations. It provides an overview of real-time parking occupancy, lists of active and upcoming reservations, vehicle entry/exit events, and alerts for policy violations (e.g., repeated late departures or misuse of priority carpool spaces). Staff can conceptually approve

or reject parking access, trigger manual overrides when necessary, and inspect audit records to support incident analysis.

In line with the course emphasis on conceptual and user-centred design, these interfaces are described in terms of screens, navigation and information flows only; no HTML, CSS or implementation details are specified in this workshop.

3.3 Conceptual Application Services

The Application Layer refines the services initially proposed in Workshop 2, specialising them for parking and exit-time carpooling:

- Authentication Service – manages institutional login, restricting access to verified university users (e.g., @udistrital.edu.co), and exposes role information (student, staff, admin) to other services.
- Parking Inventory Service – maintains the conceptual state of each parking space (available, reserved, occupied, carpool-priority, blocked) per campus and zone, and responds to queries from the student and security interfaces.
- Parking Allocation and Policy Engine – applies configurable rules to decide whether to grant or deny parking requests, prioritise shared vehicles, enforce maximum parking times and flag repeated misuse for later sanctions.
- Exit Carpool Matching Service – registers offers and requests for shared trips leaving the campus, and matches drivers and passengers by destination zone, departure window and available capacity.

- Notification and Messaging Service – sends confirmations and updates for parking reservations, carpool matches and cancellations, and supports basic communication between matched drivers and passengers.

These services are specified as conceptual components with well-defined responsibilities and interactions, providing a blueprint for future implementation choices.

3.4 Conceptual Fault-Tolerance and Data Resilience

To address the risk of last-minute driver cancellations and to maintain consistent data under peak load, the architecture incorporates fault-tolerance and resilience patterns at a conceptual level:

- Event-Driven Cancellation Handling.

When a driver cancels shortly before departure, the Exit Carpool Matching Service emits a `DriverCancellationEvent` that triggers an immediate search for alternative carpool offers to the same or nearby city zones within a compatible time range. If a suitable driver is found, passengers are reassigned and notified; if not, they are informed and offered later departure options or institutional information on public transport, without relying on bike-sharing.

- Consistent and Auditable Data.

Parking reservations, occupancy updates and carpool matches are required to be recorded as atomic, consistent operations in the conceptual data model, avoiding partial updates that could misrepresent occupancy or matches. All critical actions—such as policy overrides and reputation changes due to cancellations—are logged in audit records, which support monitoring and future analysis.

These resilience properties define how the system is expected to behave under real-world conditions and guide the selection of concrete technologies in a future development phase, while keeping the Workshop 3 deliverable strictly at the level of conceptual architecture.

4. Risk Management

Para garantizar la viabilidad y robustez del sistema en la sede de la Calle 40, se realiza una identificación y análisis sistemático de riesgos. Los riesgos se clasifican en cuatro categorías fundamentales

	Category	Risk Description	Impact / Probability	Mitigation Strategy	Contingency Plan
P1	Technical	Data Desynchronization: The app displays available parking slots at Calle 40 campus when it is actually full.	High / Medium	Implement real-time communication protocols to ensure instant data updates.	Shift immediately to a manual logbook managed by campus security guards to prevent traffic jams at the entrance.

P2	Technical	Chat Service Downtime: The messaging platform crashes during peak morning or evening hours due to high traffic volume.	High / Low	implement message rate limiting per user.	Enable a static, "bulletin board" for basic text announcements while the messaging server is restarted.
P3	Security	Unauthorized Users / Infiltrators: Non-university individuals access the platform, creating spam or safety concerns for students.	Very High / Low	Enforce mandatory user registration and login exclusively via institutional email accounts (@udistrital.edu.co) .	Terminate the suspicious session, block the user's IP and account, and alert the system administrator to audit the security breach.
P4	Security	Data Breach / Leak: Sensitive user information (phone numbers, names, or license plates) is publicly exposed.	Very High / Low	Mask phone numbers within the chat interface and encrypt all sensitive data at rest in the database.	Put the application into "Maintenance Mode" (offline) to patch the vulnerability and immediately notify affected users.

P5	Operational	Low User Adoption: Drivers prefer to stick with old, informal WhatsApp groups instead of migrating to the new app.	Medium / Medium	Design an intuitive User Interface (UI) that allows publishing an available carpooling slot in less than 3 clicks.	Partner with Student Welfare or the Campus Council to provide physical incentives, such as parking space priority.
P6	Operational	Spam and Fake Carpooling Routes: Users post chat messages containing misinformation, sales, or non-existent trips.	Low / Medium	Integrate a community-driven "Report Message / User" button for quick flagging.	Grant temporary moderator privileges to campus security or the project team to delete fake posts with a single click.
P7	Management	Scope Creep (Losing Focus): Try to add complex features that affect the project.	High / High	Freeze complex system requirements and strictly adhere to a Minimum Viable Product: parking management and basic chat.	Remove any failing secondary features immediately and deliver only the core functional modules..

TECHNICAL RISKS

- Real-time desynchronization of the parking lot status.

Description: The system displays that there are available slots in the Calle 40 parking lot when it is actually full, due to server latency or failures in the entry/exit registry.

Mitigation: the parking lot status must be updated immediately, along with a reconnection and automatic update mechanism on the client side in case of cellular signal loss within the campus.

Contingency Plan (Plan B): If the servers fail, the parking lot security guards close the digital registry and temporarily return to manual control with a physical logbook so as not to generate traffic jams at the campus entrance.

- Saturation or crash of the chat service during peak hours.

- Description: High traffic of simultaneous messages during university peak hours that causes the messaging server to crash.
- Mitigation: Design the backend using asynchronous threads for message handling and apply request limiting policies to avoid server overload.
- Contingency Plan (Plan B): Enable an official and static alternative channel in the app (a simple text "bulletin board" without interactive chat) that consumes very little data while the team restarts the messaging server.

SECURITY RISKS

-Identity theft of drivers or passengers.

Description: Individuals outside the Universidad Distrital or malicious users who access the chat to post false information or put the safety of students at risk.

Mitigation: Restrict registration and access to the platform layout obligatorily using exclusively the institutional email (@udistrital.edu.co) and/or student code.

Contingency Plan (Plan B): If an infiltrated user is detected, the system immediately blocks their IP and account, and sends an alert to the system administrator to review how they bypassed the security of the institutional email.

-Exposure of personal data in the chat.

Description: Sensitive information such as cell phone number, vehicle license plate, or private routes of drivers is exposed publicly in an insecure manner.

Mitigation: protect sensitive data in the database and allow the chat to hide the phone number, managing internal communication solely through verified institutional usernames.

Contingency Plan (Plan B): put the system in "Maintenance Mode" and notify affected users while the database vulnerability is patched.

OPERATIONAL RISKS

-Low adoption of the platform by drivers.

Description: Drivers continue to prefer using the existing informal WhatsApp groups and do not post their available slots in the new application.

Mitigation: Design an extremely simple user interface (UI) where publishing a slot takes less than 3 clicks, and incentivize usage through benefits in the allocation priority of physical parking slots.

Contingency Plan (Plan B): Partner with the Student Council or University Welfare of the Calle 40 campus to launch a physical dissemination campaign (posters in the parking lot, publications on institutional networks) offering incentives, such as parking priority to those who use the app.

-Fake reports or spam in the routes chat.

Description: Users who post fake routes or saturate the chat with messages that do not correspond to campus mobility.

Mitigation: Implement a community moderation module (a "Report user/message" button) that temporarily blocks users who accumulate three negative reports.

Contingency Plan (Plan B): If the automatic filter fails, "moderator" permissions are given to the security guards themselves or to a couple of students from the project so they can delete offensive or false messages in real time with a single button.

PROJECT MANAGEMENT

- Loss of focus

description: Adding complex features to the app that are not viable and harm the system

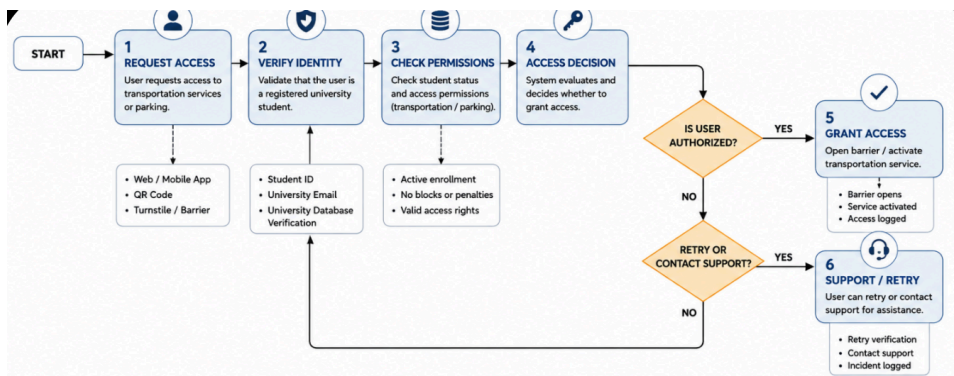
Contingency Plan (Plan B): If the system is harmed by trying to add extra things, those additional functions are completely eliminated and the functional base of the project is delivered.

5. Quality Attributes & Metrics

To have a clear validation of the system and measurable technical operational compliance, the following specific quality criteria have been established:

- **Performance Efficiency:** Metric: The project should process requests in a response time of no more than 5 seconds under a simultaneous load of up to 500 requests during peak hours (07:50,9:50,11:50,13:50)

- **Validation Approach:** it is necessary to implement something that manages that those who use these transportation services and those who access the parking spaces are really students of the university, since it is a big risk that users who are not part of the university can enter the parking lot and "sneak" by taking spaces from those who do need them.
- **Reliability (availability):** it must function properly and that errors do not occur where registered users have been denied entry or vice versa.
- **Security (data confidentiality):** Metric: the information must be completely confidential, risks must be avoided, although if users do not wish to provide information, nothing strange will be necessary, making this issue even easier in this project.



6. Project management plan

6.1 Project Charter

- Project Name: Campus parking and exit carpool coordination.

- Main objective: Establish a highly secure, scalable vehicle sharing and parking space management system, solving poor communications practices and disorder within the engineering faculty parking lot.
- Scope of the project: Delivery of a system covering the topics seen throughout the semester with a modular structure, a management register, operational risks, validation tests and an appropriate system design.
- Project Actors: Engineering Students from the District University, Faculty of Engineering.

6.2 Team structure and responsibility framework (RACI matrix)

Project leadership and responsibilities have been randomly and strategically distributed across our team to ensure compliance with software design life cycles (SDLC). This distribution is evidenced through a Responsibility Assignment Matrix (this is an example assignment and assumption of what it would be like in a large and business project, in this case we all did a little of each thing):

- Project Manager: Miguel Mateo Guillén Guzmán
- Lead Software Architect: Nicolás Acero Ladino
- Data and Backend Engineer: Julián David Muñoz Revelo
- QA & Frontend Engineer: Santiago Alexander García Bermeo

Project Phase / Key Deliverable	Miguel Mateo Guillén Guzmán	Nicolás Acero Ladino	Julián David Muñoz Revelo	Santiago Alexander García Bermeo
Architecture Refinement & Microservices	I	R	A	C
Risk Matrix & Quality Metrics Plan	A	R	C	C
Backend Service Development & APIs	C	C	R	I
Frontend UI Prototype Design	I	I	C	R
End-to-End QA Integration Testing	A	C	R	R

Where: R = Responsible, A = Accountable, C = Consulted, I = Informed.

6.3 Project schedule

The project will be developed in a structured 4-week calendar.

- **Week 1:** Generation of architecture and risk profile study

Deliverables: technical specification documentation, formalized ISO/IEC 25010 definition, component designs and modeling.

- **Week 2:** Data engineering and core backend infrastructure

Deliverables: development of secure provisioning of relational database schemas.

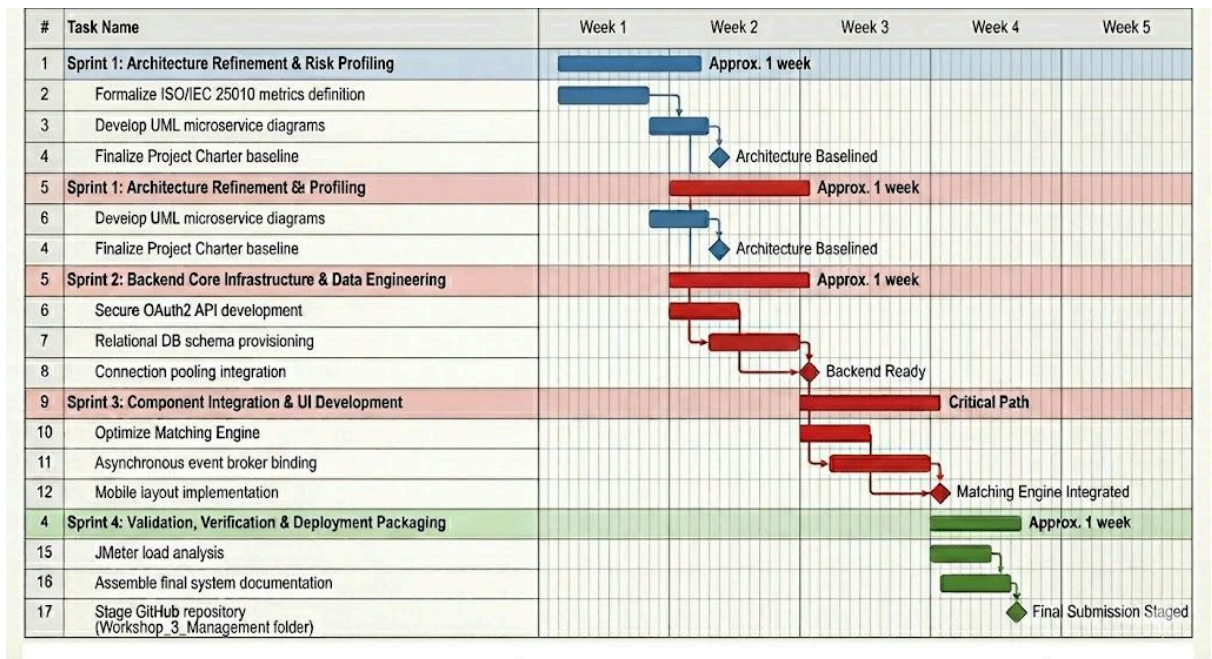
- **Week 3:** Algorithmic integration and user interface development

Deliverables: Algorithms code, front-end visual interface style.

- **Week 4:** Quality control testing, validation and repository implementation.

Deliverables: Performance testing, creating validation checklists, formatting the final submission structure within the Workshop_3_ repository folder.

The next diagram is a representation of what the calendar and planning process could be like for a similar but much larger project at a professional and business level. This seeks to understand how companies work and make a relationship between what we do with this work and how we can compare or prepare ourselves in the future for business situations.



7. Implementation Strategy

This section details the strategic plan to deploy the parking management and chat system at the Calle 40 campus, minimizing operational risks and ensuring a smooth transition for the university community.

7.1 Phased Deployment Approach

To mitigate critical failures in production, the launch of the system will be structured into three consecutive phases:

Phase 1: Alpha Testing and Infrastructure Validation: The system will be deployed in a controlled testing environment. The development team and a small group of

students will simulate the publication of transportation slots and parking reservations to verify chat speed and data synchronization.

Phase 2: Closed Pilot Plan: The application will be implemented exclusively with the security personnel (guards) of Calle 40 and a group of 30 regular drivers (faculty and students). This will allow evaluating the system's performance against cellular signal loss in certain areas of the parking lot and calibrating the ease of use of the interface.

Phase 3: Official Launch: After correcting the errors detected in the pilot, the platform will be opened to the entire community of the faculty of engineering at the Calle 40 campus.

7.2. Technical Infrastructure and Prerequisites

The deployment of the solution requires compliance with the following specifications and technological dependencies:

Server and Database: Cloud server (or local institutional server) with support for software execution environments and a relational database configured to securely store parking access records and credentials.

Real-Time Protocol: Configuration to ensure that chat messages and parking slot availability are updated immediately without saturating the main server.

Security Integration: Enabling and permissions of the institutional authentication service of Universidad Distrital to mandatorily validate @udistrital.edu.co accounts before allowing entry to the app.

Campus Access: Ensure that security guards have a mobile device or a computer station at the parking lot entrance with stable access to the university's internal network.

Change Management and User Adoption

The transition from manual control and informal WhatsApp groups toward the new platform will be managed through the following strategies:

Security Personnel Training: Practical sessions will be held with the campus guards to teach them how to use the parking control module, manage the physical contingency plan (in case of network failures), and use the chat moderation button.

Institutional Dissemination Campaign: In coordination with University Welfare and the communication channels of the Faculty of Engineering, informative posters with QR codes will be installed at the entrances of the Calle 40 parking lot to facilitate the direct download of the application.

Adoption Incentive Strategy: A temporary benefit of priority allocation of physical spaces in the parking lot or agility in the entrance line will be managed with the campus administration for those drivers who actively publish their available slots in the system's chat.

8. Conclusión

System Lifecycle Integration and Maturation: Moving from the empirical identification of campus mobility challenges (Workshop 1) and conceptual architectural design (Workshop 2) to management, quality, and risk planning (Workshop 3) successfully consolidated the transition from an abstract idea into a mature, realistic technological solution.

Operational Feasibility: Redefining the project scope proved that technical feasibility and real-world environmental constraints must guide the design. This new focus drastically reduced architectural friction, lowered cloud infrastructure costs, and optimized development resources toward the most immediate pain points of the university community.

System Robustness: Refining the system under engineering principles demonstrated that robust software goes further. The systematic mitigation of technical and security risks (such as data synchronization through [WebSockets] and strict restriction with institutional authentication), complemented by operational contingency plans for security personnel, guarantees service continuity against any infrastructure failure.

Project Management: Incorporating formal project management methodologies (team roles structuring, critical path schedule analysis, and scope creep control) proved to be indispensable. These tools ensure that the mobility system can be implemented within defined time and budget constraints, mitigating uncertainty and securing a coordinated community adoption alongside University Welfare and campus administration.